

Hybrid Physics + Machine Learning for Optimizing High-Voltage Impulse Generator Settings

Problem Overview

High-voltage insulators used in power transmission systems must undergo impulse testing to verify their ability to withstand lightning strikes and switching surges. During these tests, engineers use a multi-stage impulse generator to produce standard lightning and switching impulse waveforms.

A major challenge is selecting the correct generator settings. The required resistor values, charging voltages, and stage configurations depend on several factors such as:

- Insulator capacitance
- Measuring system capacitance
- Stray capacitances in the test setup
- Physical arrangement of the test equipment

Currently, engineers often rely on calculations followed by multiple trial runs to obtain a waveform that satisfies standard requirements. This process can be time-consuming and may require repeated adjustments.

The goal of this challenge is to develop an intelligent software tool that combines engineering calculations with machine learning to recommend suitable generator settings before physical testing begins.

Challenge Statement

Develop a software solution that predicts and recommends suitable impulse generator settings for high-voltage insulator testing.

The solution should:

- Support both lightning impulses (1.2/50 μ s) and switching impulses (250/2500 μ s).
- Estimate suitable:
 - Front resistance
 - Tail resistance
 - Charging voltage
 - Number of active generator stages
- Predict expected waveform characteristics.
- Verify whether predicted waveforms satisfy specified tolerance limits.
- Rank multiple feasible settings and recommend the best options.
- Provide engineers with a reliable starting point before conducting actual tests.

Expected Solution

Participants are expected to build a software proof-of-concept, preferably a web application.

The system should include:

1. Engineering Calculation Module

Uses circuit equations and input parameters to identify physically feasible operating conditions.

2. Machine Learning Engine

Learns the relationship between:

- Generator settings
- Load characteristics
- Waveform outputs

and predicts the resulting impulse waveform parameters.

3. Recommendation Engine

Suggests the most appropriate generator settings and ranks them based on performance and compliance.

4. Visualization Dashboard

Displays:

- Recommended settings
- Predicted waveform parameters
- Compliance status
- Waveform comparison plots
- Performance metrics

Inputs to the System

The user should be able to specify:

- Impulse type (Lightning / Switching)
- Desired crest voltage
- Insulator capacitance
- Measurement-system capacitance
- Estimated stray capacitance
- Available front resistor values
- Available tail resistor values

Expected Outputs

The solution should provide:

Recommended Settings

- Number of generator stages
- Charging voltage per stage
- Front resistance
- Tail resistance

Predicted Waveform Characteristics

- Front time
- Peak time (for switching impulses)
- Tail time
- Crest voltage

Compliance Information

- Percentage deviation from target values
- Within-limit / Outside-limit indication
- Ranking of alternative solutions

Visualization

- Target waveform vs Predicted waveform
- Summary dashboard for decision-making

Suggested Technical Approach

Teams may consider the following workflow:

1. Generate or use the provided dataset.
2. Perform exploratory data analysis.
3. Develop physics-based calculations for waveform estimation.
4. Train machine-learning models to predict waveform parameters.
5. Compare multiple algorithms.
6. Implement a recommendation engine.
7. Create a user-friendly dashboard.
8. Validate recommendations against tolerance requirements.

Additional Guidance:

In practical impulse testing, the electrical parameters of the test circuit are not fixed. The test-object/load capacitance, measuring-system/divider capacitance, stray capacitance, stray/connection inductance, connecting leads, physical arrangement and grounding/return-path configuration may vary depending on the test object and test setup. These variations can significantly influence the generated impulse waveform and, consequently, the optimum generator settings.

Therefore, the proposed software should consider these variable and configuration-dependent parameters while recommending the generator configuration.

In addition, the optimization should consider the actual available impulse-generator parameters and operating constraints, including:

- Maximum charging voltage
- Number of available stages
- Stage voltage
- Stored energy per stage and total stored energy
- Impulse capacitance per stage
- Available front resistors
- Tail resistors for lightning and switching impulses
- Charging resistor
- External front/tail resistors, wherever applicable
- Potential and discharging resistors
- Basic/load capacitance of the generator
- Sphere-gap dimensions and configuration
- Available generator connection/configuration possibilities

For our impulse generator, the software should support the required impulse test voltage range of approximately 50 kV to 2400 kV, with a minimum of two active generator stages, subject to the applicable generator configuration and operating limits.

It would also be important to use actual measured impulse-waveform data from different test objects and test configurations for calibration and validation of the ML model, in addition to the physics-based calculations. This will help improve the accuracy and reliability of the recommended settings under actual laboratory conditions.

The figure below would help distinguish between fixed generator parameters/constraints and variable test-circuit parameters, such as test-object capacitance, measuring-system capacitance, stray capacitance, stray inductance and physical test configuration.

Parameters from HV Impulse Generator (e.g., HAEFELY 508085/E) to be considered in the Software for Generator Setting Optimization

A. Generator Technical Data (Mandatory Inputs/Constraints)

1	Maximum charging voltage (ΣU)	: 2400 kV
2	Maximum stored energy at rated voltage (W)	: 30 kJ
3	Number of stages (B)	: 12
4	Stage voltage (U_s)	: 200 kV
5	Stored energy per stage (W_s)	: 2.5 kJ
6	Impulse capacitance per stage (C_s)	: 0.125 μ F
7	Wave form according to standard	: 1.2/50 μ s; 250/2500 μ s
8	Front resistor per stage (R_{s1})	: 30 Ω , 465 Ω , 3.7 k Ω (brown / blue / yellow)
9	Tail resistor per stage for lightning (R_{p11})	: 520 Ω (grey)
10	Tail resistor per stage for switching (R_{p12})	: 22 k Ω (green)
11	Charging resistor per stage (R_L)	: 135 k Ω (Lilac)
12	External front resistors (R_{se1} , R_{se2} , R_{se3})	: -

B. Additional Generator Elements

13	External tail resistor (R_{pe})	: -
14	Potential resistor (R_{pot})	: 2 M Ω (grey)
15	Discharging resistor (R_{erd1})	: 5.45 k Ω (grey)
16	Discharging resistor (R_{erd2})	: 13 k Ω (grey)
17	Sequence of pulses (k)	: 2 / min

C. Test Circuit and Mechanical Data

18	Basic load capacitance (C_{bo})	: 545 pF
19	Sphere diameter (for sphere gap)	: 25 cm
20	Connection possibilities	: e.g., 1 p12 s; p s
21	Total height of generator	: approx. 8.7 m
22	Base surface of generator ($A \times B$)	: -
23	Total weight (G)	: -
24	Maximum weight of heaviest part (g)	: -

D. Test Requirements (from user input)

25	Impulse type	: Lightning (1.2/50 μ s) / Switching (250/2500 μ s)
26	Required test voltage (crest)	: 50 kV to 2400 kV
27	Test object capacitance	: (variable)
28	Measuring system / divider capacitance	: (variable)
29	Estimated stray capacitance	: (variable)
30	Stray loop inductance	: (variable)
31	Available front resistor values	: (from generator inventory)
32	Available tail resistor values	: (from generator inventory)

HAEFELY 508085/E		1-1
1. Technical data		
Max. charging voltage	ΣU	= 2400 kV
Max. stored energy at rated voltage	W	= 30 kJ
Number of stages	B	= 12
Stage voltage	U_s	= 200 kV
Stored energy per stage	W_s	= 2.5 kJ
Impulse capacitance per stage	C_s	= 0.125 μ F
Wave form according to IEC		1,2/50 μ s; 250/2500 μ s
Front resistor per stage	R_{s1}	= 30 Ω , 465 Ω , 3.7 k Ω (brown / blue / yellow)
Tail resistor per stage for lightning impulses	R_{p11}	= 520 Ω
Tail resistor per stage for switching impulses	R_{p12}	= 22 k Ω (green)
Charging resistor per stage	R_L	= 135 k Ω (lilac)
External front resistor	R_{se1}	= -
External front resistor	R_{se2}	= -
External front resistor	R_{se3}	= -
External tail resistor	R_{pe}	= -
Potential resistor	R_{pot}	= 2 M Ω (grey)
Discharging resistor	R_{erd1}	= 5.45 k Ω (grey)
Discharging resistor	R_{erd2}	= 13 k Ω (grey)
Sequence of pulses	k	= 2 / min
Basic load capacitance	C_{bo}	= 545 pF
Sphere diameter		= 25 cm
Connection possibilities: 1 p12 s; p s.		
Total height of generator	H approx.	= 8.7 m
Base surface of generator	A x B	= -
Total weight	G	= - kg
Max. weight of heaviest part	g	= - kg

Important Constraints for the Software Tool

- 1 The software shall consider all the above generator parameters in the optimization process.
- 2 The recommended configuration shall be within the practical operating limits of the impulse generator.
- 3 Impulse test voltage range to be supported: 50 kV to 2400 kV, with a minimum of 2 stages.